Natural Computing

Report on Assignment 3

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Group 28

1 Part A

1.1 Reproducing quantization

1.2 Finding optimal color palette

1.3 Results

2 Part B

2.1 Introduction

The Boids model from Craig Reynolds is one of the most popular models of swarm intelligence and group behavior. It uses boids, which are bird like objects, that follow 3 simple rules: alignment, cohesion, and separation. These simple rules govern the interactions between individual boids, which gives rise to a more complex emergent behaviors such as flocking. This is also known as swarm intelligence. Swarm intelligence is some collective behavior that emerges when individual agents interact with each other. This can lead to flocking, fish schools, and insect swarms. The goal of this project is to implement the Boids model and explore how the swarm behavior emerges from simple rules and how modifying the parameters affect it.

— Just some info —

Explain how the implementation satisfies all the requirements - fixed speed...

The model keeps Boid speeds fixed; agents should update their direction but move at a constant speed v which is a parameter of your model. (const move = this.multiplyVector(this.dir, this.speed))

Explicitly considers what should happen at the boundary of your simulation field: The boundary is wrapped. When a boid exits from one side, it reappears on the opposite side.

uses two interaction radii: one for separation, and one for cohesion/alignment. (Ask yourself: which one should be smaller and why?): The inner radius is used for separation, while the outer radius is used for cohesion and alignment. The inner radius should be smaller than the outer radius because separation requires a closer distance to trigger the behavior, while cohesion and alignment can occur at a larger distance.

excludes the boid itself from its own neighborhood when calculating the update rules, so it does not try to separate, align or cohere to itself: The implementation excludes the boid itself from its own neighborhood when calculating the update rules. This is done by ensuring that when iterating through the neighboring boids, the current boid is not included in the calculations for separation, alignment, and cohesion. (if(p.id == x.id) continue)

2.2 Methods

The JavaScript framework was used to implement the two-dimensional Boids model. The model was defined on a square grid of 400x400 pixels with wrapped dimensions. 150 boids were randomly initialized, each moving at a constant speed. The position and direction of each boid were updated at each time step based on the three rules. The simulation was ran for 300 time steps and the used parameters in this project are displayed in Table 1 below.

Boids follow three simple interaction rules: alignment, cohesion, and separation. Alignment controls how each boid moves towards the average direction of the other boids nearby. Similarly, cohesion controls how quickly each boid tries to move towards the average position of the other boids around. Separation dictates how much boids try to keep a certain distance from each other to avoid collision. Together, these rules dictate how boids interact with each other. However, other than these 3 basic behavior rules, each boid has an inner and outer radius. The outer radius is associated with cohesion and alignment, while the inner radius is linked to separation. The outer radius is the boids' visual and interaction range. It is the distance within which the boid can see other boids and interact with them, and is visualized as a gray circle surrounding the boid. The inner radius, depicted by a red circle around the boid, prevents collision and is the opposite of the interaction range. If a boid enters another boids inner radius, they are repelled away, controlled by the separation weight. Obviously, the outer radius should be larger than the inner radius so cohesion and alignment can occur.

Parameter	**Value**
$Radius_{\text{inner}}$	10
$Radius_{\text{outer}}$	25
Alignment weight	1
Coherence weight	1
Separation weight	1
Number of boids	150

Table 1: Parameters.

To analyze the swarm, the order and nearest-neighbor quantitative metrics were used. Order parameter is the average normalized velocity of the Boids. Order parameter tells us to what degree the swarms align in their directions, and the output is squashed between 0 and 1, 0 meaning completely random directions and 1 meaning perfect alignment. The nearest-neighbor distance is the average nearest-neighbor distance across all boids. It measures the total average distance across all boids and can tell us when individual boids get separated from the swarm or when individuals get too close to each other.

2.3 Results

Visuals. Quantitative results. Parameter experiments.

Is there swarm alignment? How does alignment change over time? How do boids maintain spacing? How do parameters influence behavior?

Time	**0**	**100**	**200**	**300**
Order parameter	0.0208	0.0753	0.105	0.616
Nearest-neighbor distance	13.956	5.925	4.278	3.405

Table 2: Quantitative results for default parameters.

Time	**0**	**100**	**200**	**300**
Order parameter	0.096	0.509	0.751	0.695
Nearest-neighbor distance	14.466	9.076	8.053	7.506

Table 3: Quantitative results for alternate parameters.

It is important to note that due to the random initializations the results can greatly vary between runs.

It is possible to reach high alignment in 300 steps with these parameters.

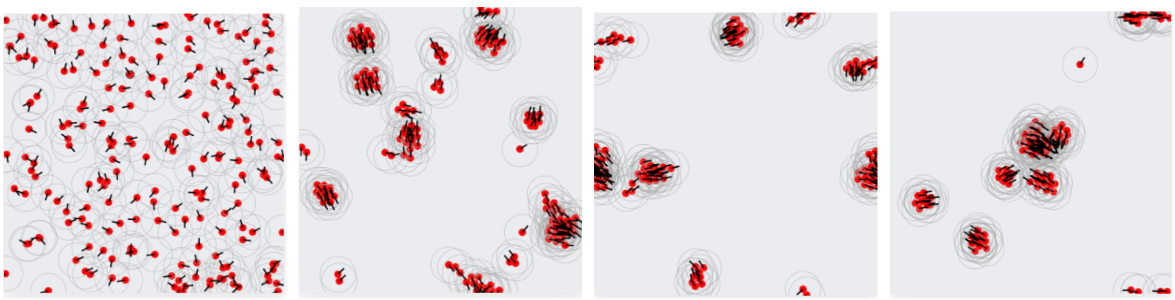
The visualizations in Figure 1 show the positions of the boids at different time steps. Four snapshots were taken at time steps 0, 100, 200, and 300, and are separated into an initial state, final state, and two intermediate states.

Red color represents the inner radius, while the black bar coming out of it indicates in which direction each boid is facing. The outer radius is shown as a gray circle and is each boids' interaction range.

150 boids were initialized randomly on a 400x400 field with wrapped boundaries. Boids then move around, updating their positions and orientations based on the alignment, coherence, and separation rules. Boids interact with each other by coming in each other interaction range, forming flocks as they align and cohere with each other, while also maintaining a certain distance to avoid collisions. Over time, larger flocks form as boids and smaller flocks join together. However, as shown in Figure 1d, not all boids are able to find a flock by time step 300 and remain isolated until they bump into another boid or flock.

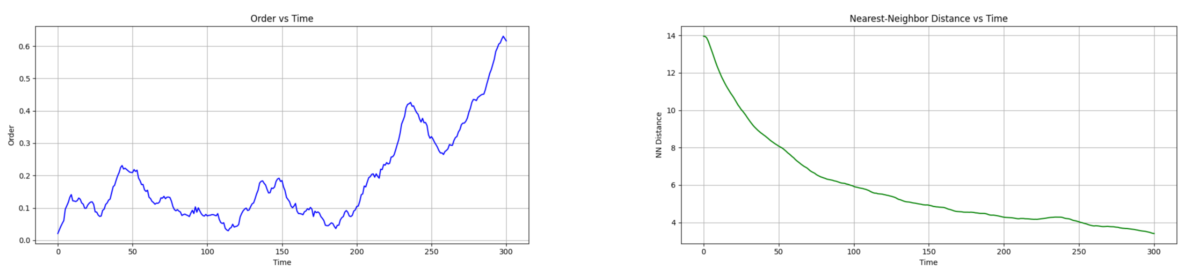
The boids are distributed across the simulation field, and their positions and orientations change over time as they interact with each other based on the alignment, coherence, and separation rules.

Over time, we can observe the formation of flocks as the boids align and cohere with each other. The separation behavior can also be seen as boids maintain a certain distance from each other to avoid collisions.



(a) Time: 0 steps (b) Time: 100 steps (c) Time: 200 steps (d) Time: 300 steps

Figure 1: Visualizations of boids at four different time steps using default parameters.



(a) Order over time (b) Nearest-neighbor distance over time

Figure 2: Plots of quantitative results over time.

The Table 3 below displays all the parameter configurations that were explored in order to investigate the effect of different parameters on the behavior of the system. These parameters were chosen because they allow for the exploration of all interesting configurations of the system.

By exploring different parameter configurations, interesting behaviors can be observed. By setting the coherence weight to 0, boids no longer strive to move towards the center of mass of their neighbors, instead, they align with their neighbors and keep a certain distance from them. This leads to an interesting behavior where boids form large nets that span the entire simulation field. Because of that, after a short while all boids start moving in the same direction. By increasing the outer radius, this can be sped up as boids are initialized with larger interaction radius and start forming large nets sooner. On the other hand, if the inner radius is increased to be more than half the value of the outer radius, this uniform behavior is disturbed. Ultimately, coherence weight plays a role in the sizes of the flocks that are formed, where higher values lead to larger flocks and lower values lead to tighter flocks.

Parameter	Value
$Radius_{\text{inner}}$	10, 40, 70, 100
$Radius_{\text{outer}}$	25, 50, 75, 100
Alignment weight	0, 0.5, 1
Coherence weight	0, 0.5, 1
Separation weight	0, 0.5, 1
Number of boids	100, 200, 300

Table 4: Explored parameter configurations.

Criterion	$R_{\mathbf{in}}$	$R_{\mathbf{out}}$	Align	Cohes.	Sep.	Order	NN-dist
Best Order	20	40	1	0	0	0.999	14.36
Best NN-dist	20	40	0.66	1	0	0.305	0.03
Best Ratio	5	40	0.33	1	0	0.625	0.05

Table 5: Best parameter configurations at $t = 300$ across 384 configurations (N=150, 3 repeats each). Best Order maximizes alignment, Best NN-dist minimizes average nearest-neighbor distance, and Best Ratio maximizes Order/NN-dist.

2.4 Discussion

Why behaviors emerge. How the parameters affect the behaviors. What are the limitations of the model? How to improve it?

Be sure to come to a final conclusion and discuss some potential limitations as well.

2.5 Conclusion